Backpropagation

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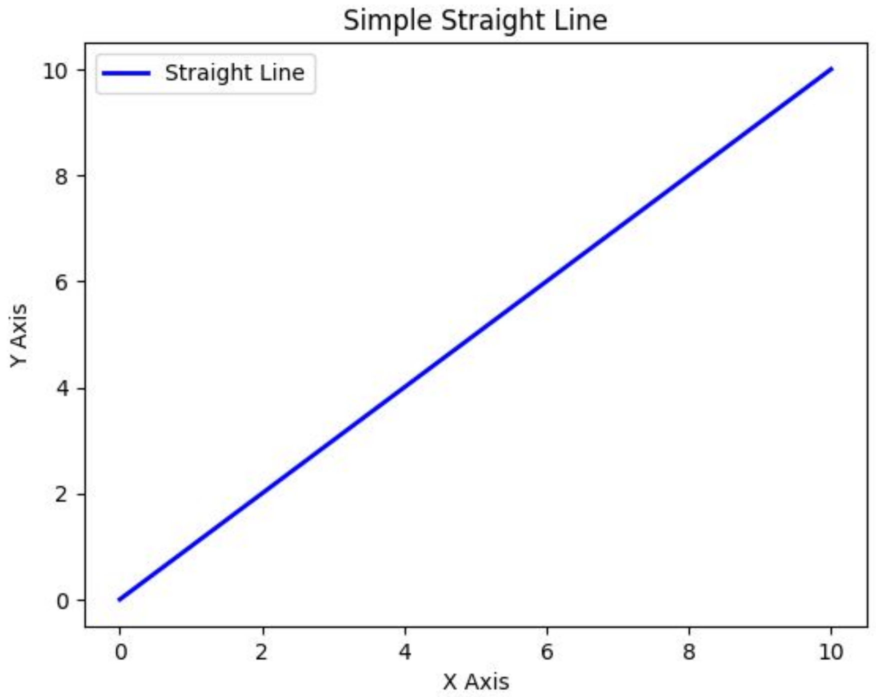
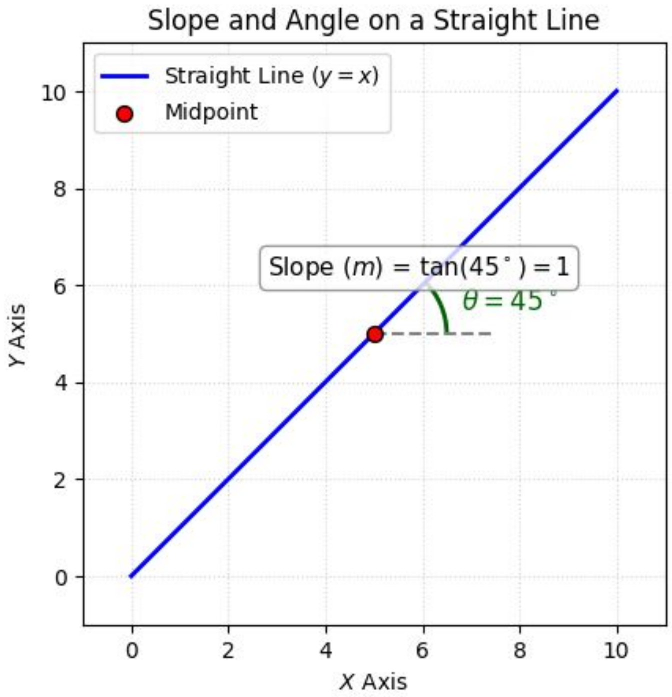
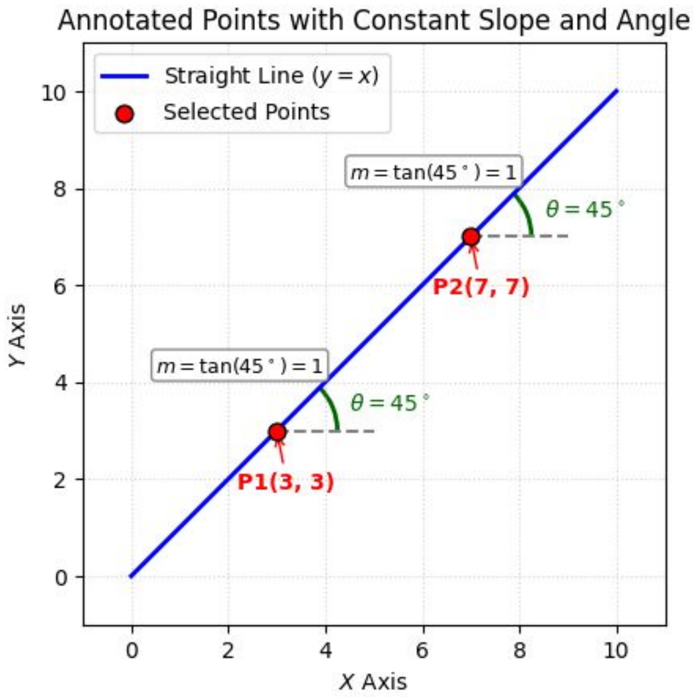
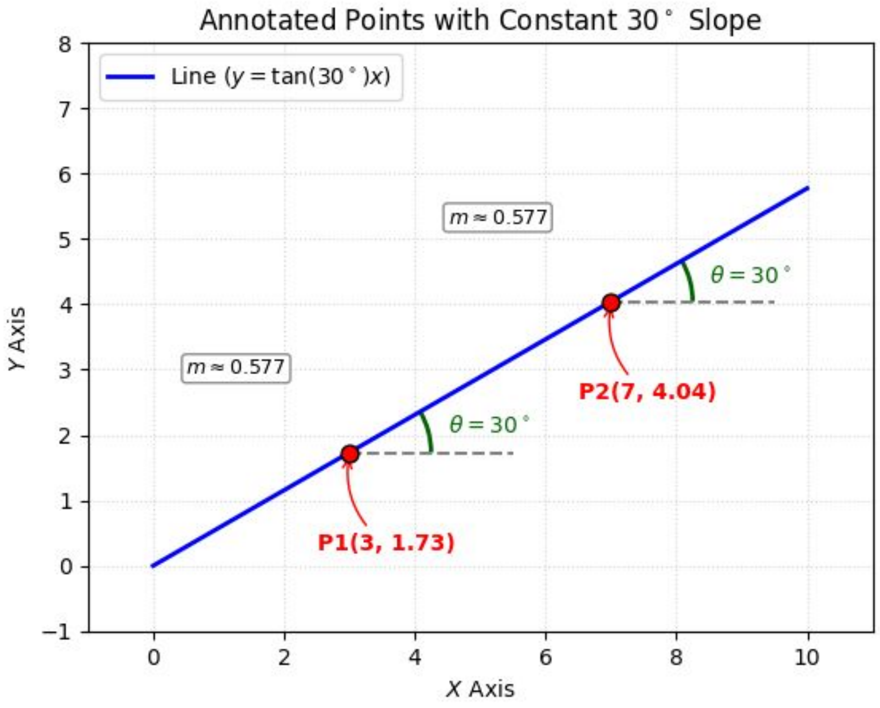
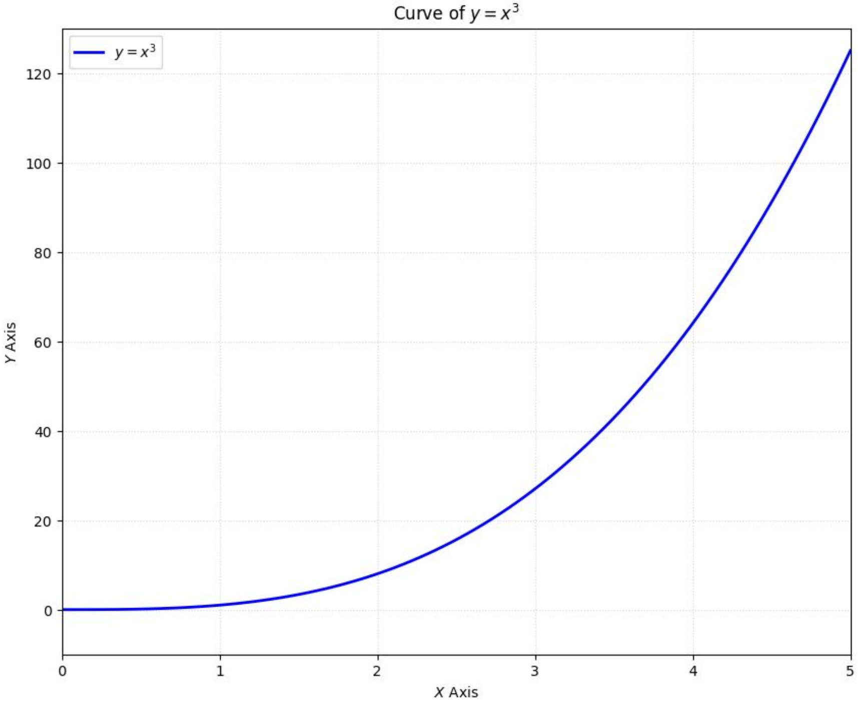

Simple Straight Line
Straight Line
10
8
6
4
2
0
Y Axis
0
2
4
6
8
10
X Axis

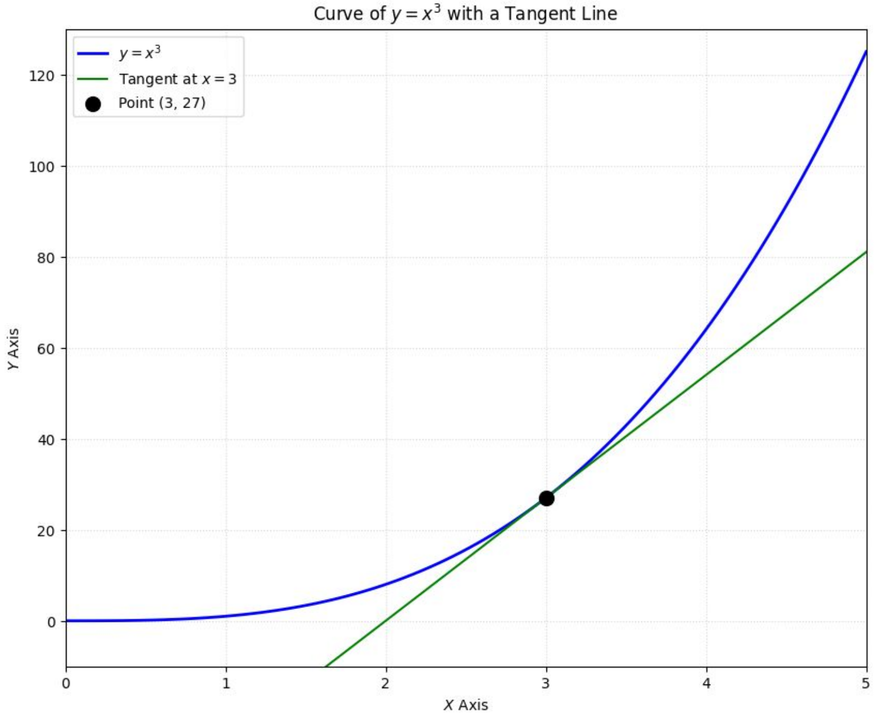

Slope and Angle on a Straight Line
Straight Line ($y = x$)
Midpoint
Slope (m) = $\tan(45^\circ) = 1$
$\theta = 45^\circ$
Y Axis
X Axis
0
2
4
6
8
10
0
2
4
6
8
10

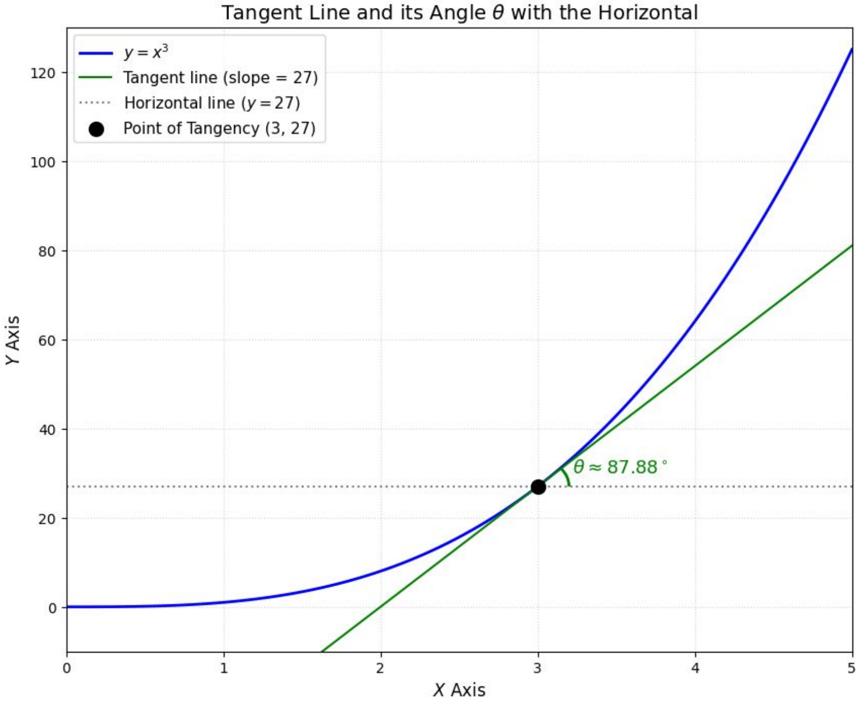


Annotated Points with Constant Slope and Angle
Straight Line ($y = x$)
Selected Points
$m = \tan(45^\circ) = 1$
$\theta = 45^\circ$
P1(3, 3)
P2(7, 7)
0
2
4
6
8
10
X Axis
Y Axis


Annotated Points with Constant 30° Slope
Line ($y = \tan(30^\circ)x$)
$m \approx 0.577$
$m \approx 0.577$
$\theta = 30^\circ$
$\theta = 30^\circ$
P1(3, 1.73)
P2(7, 4.04)
Y Axis
X Axis
8
7
6
5
4
3
2
1
0
−1
0
2
4
6
8
10

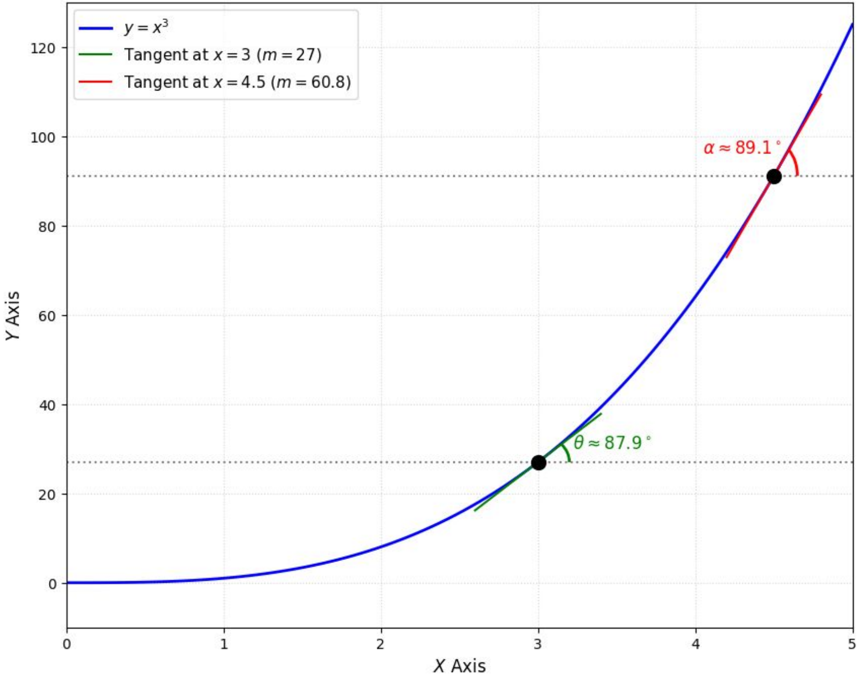

Curve of $y = x^3$
$y = x^3$
120
100
80
60
40
20
0
0
1
2
3
4
5
X Axis
Y Axis

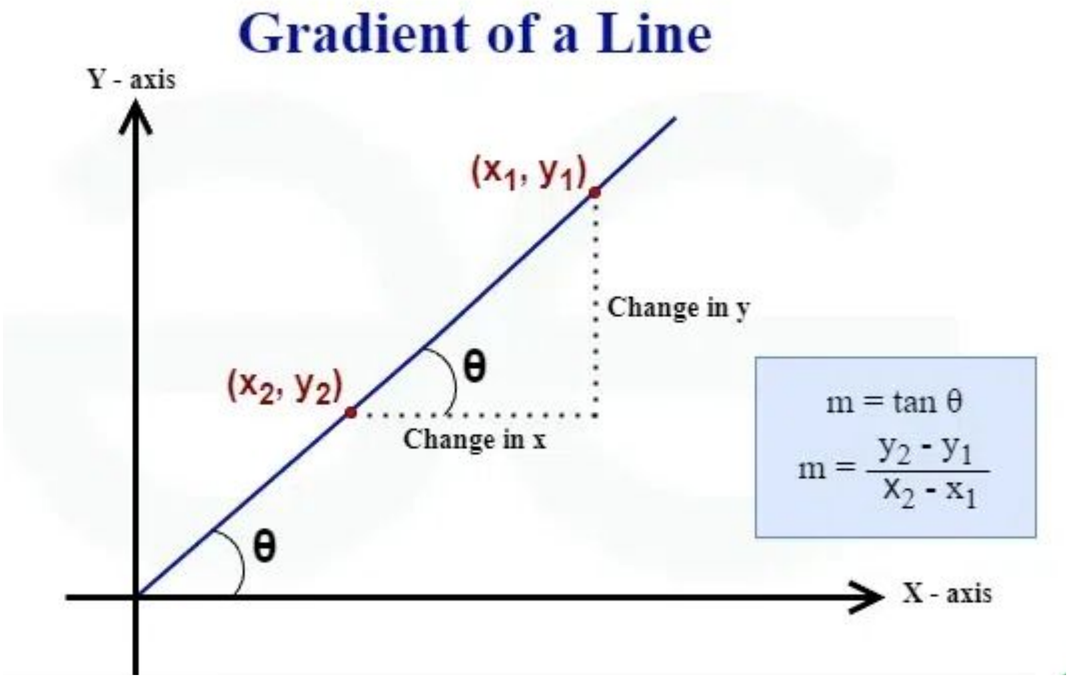

Curve of $y = x^3$ with a Tangent Line
$y = x^3$
Tangent at $x = 3$
Point (3, 27)
120
100
80
60
40
20
0
0
1
2
3
4
5
X Axis
Y Axis

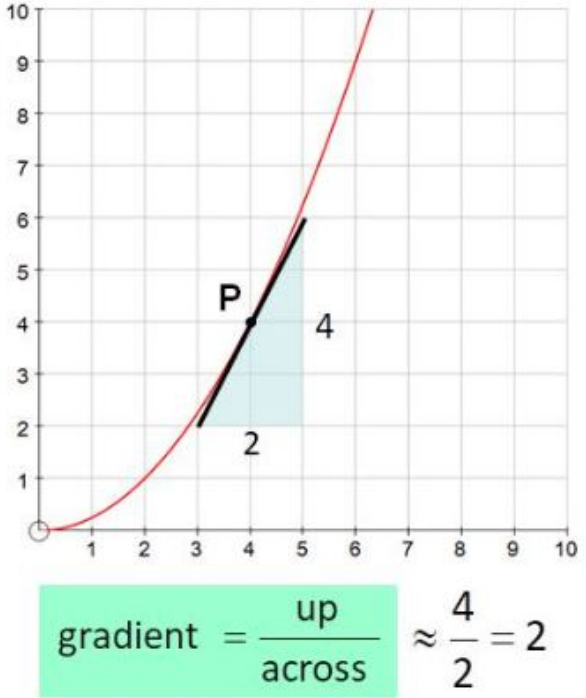


Tangent Line and its Angle θ with the Horizontal
y = x³
Tangent line (slope = 27)
Horizontal line (y = 27)
Point of Tangency (3, 27)
θ ≈ 87.88°
Y Axis
X Axis
0
20
40
60
80
100
120
0
1
2
3
4
5

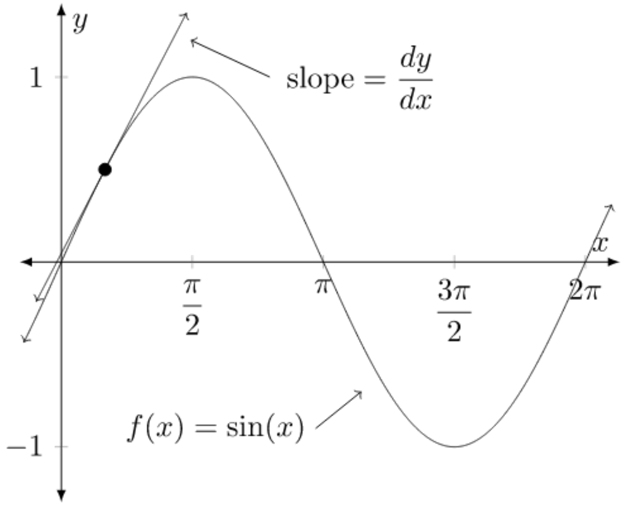
Tangent Lines and Angles at $x = 3$ and $x = 4.5$

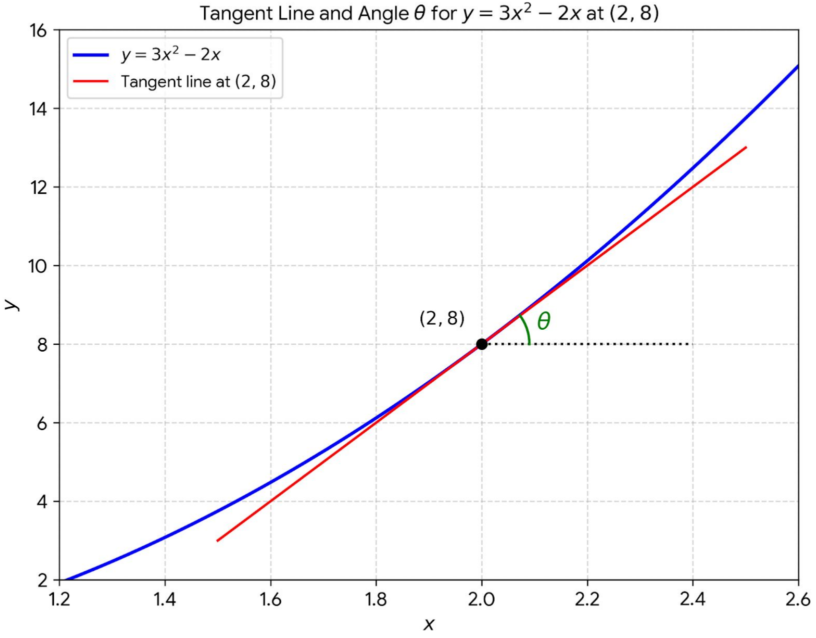



Gradient of a Line
Y - axis
(x_1, y_1)
Change in y
θ
(x_2, y_2)
Change in x
θ
X - axis
$m = \tan\theta$
$m = \frac{y_2 - y_1}{x_2 - x_1}$


10
9
8
7
6
5
4
3
2
1
P
4
2
1 2 3 4 5 6 7 8 9 10
$\text{gradient} = \frac{\text{up}}{\text{across}} \approx \frac{4}{2} = 2$

$$m = \text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x} = \frac{dy}{dx} = gradient$$





FIND THE GRADIENT OF THE CURVE $y = 3x^2 - 2x$ AT THE POINT $(2, 8)$

$$\frac{dy}{dx} = 6x - 2$$

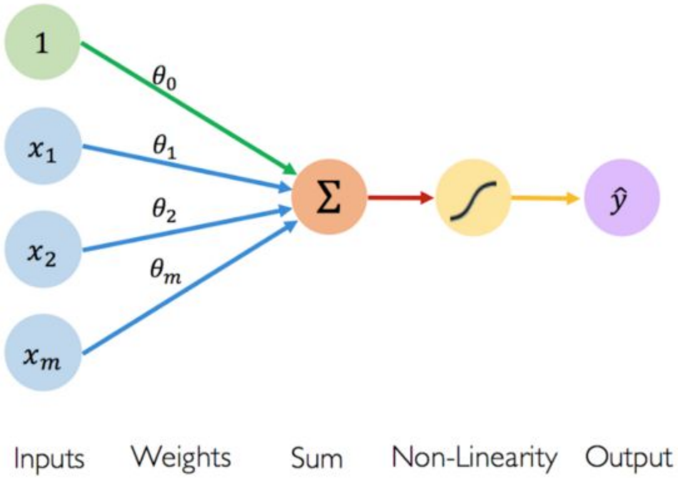
FIND THE DERIVED FUNCTION

WHEN $x = 2$, $\frac{dy}{dx} = 6 \times 2 - 2$

SUBSTITUTE IN THE x-COORDINATE

$$\frac{dy}{dx} = 10$$

THE GRADIENT AT THE POINT (2, 8) IS 10

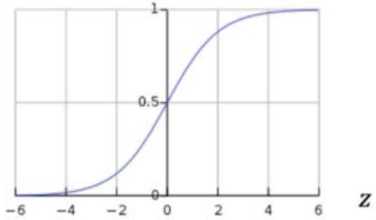


Activation Functions

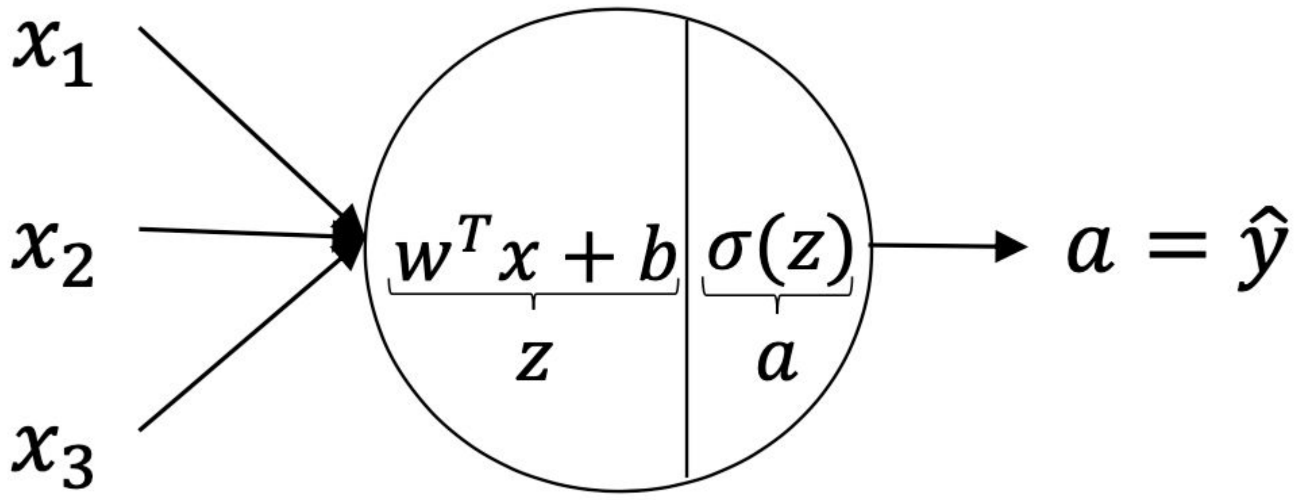
$$\hat{y} = g\left(\theta_0 + \boldsymbol{X}^T \boldsymbol{\theta} \right)$$

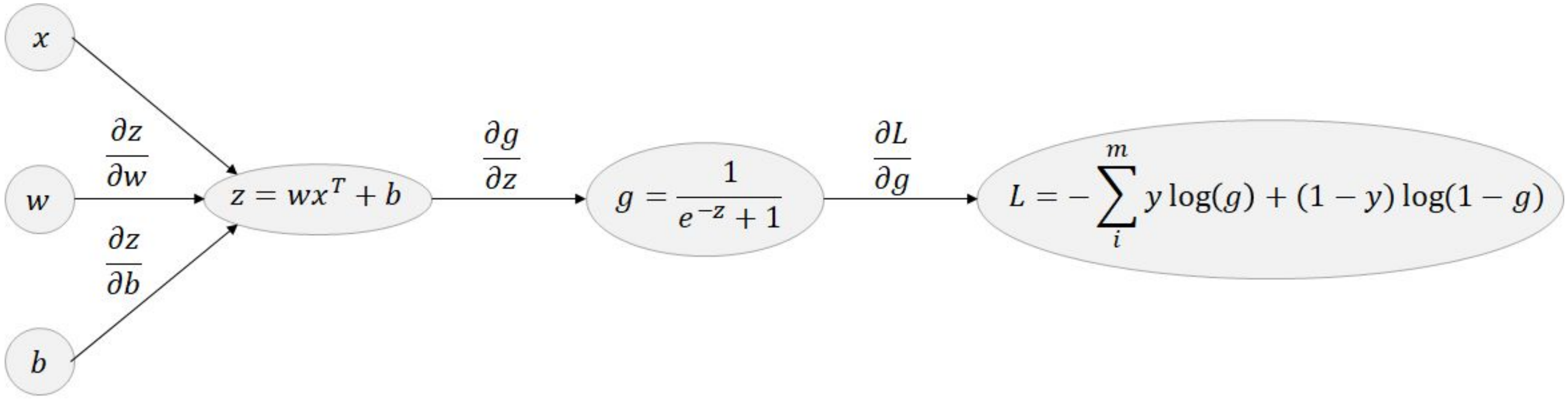
- Example: sigmoid function

$$g(z) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

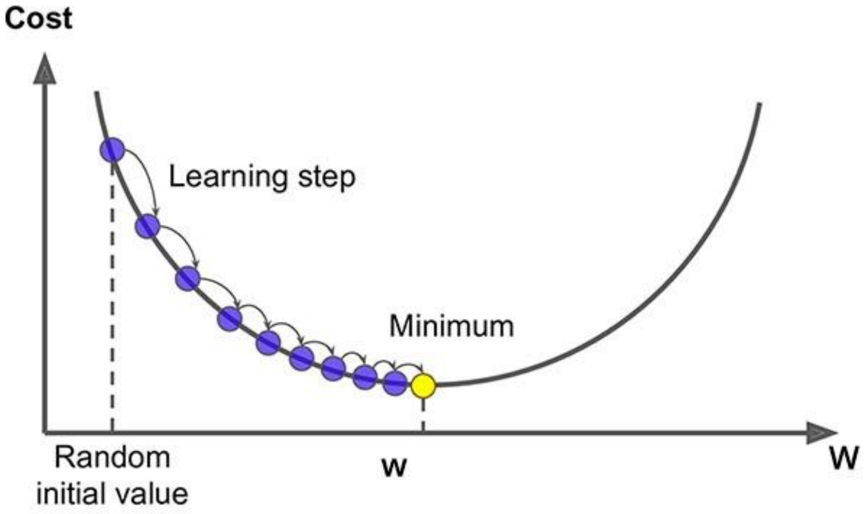


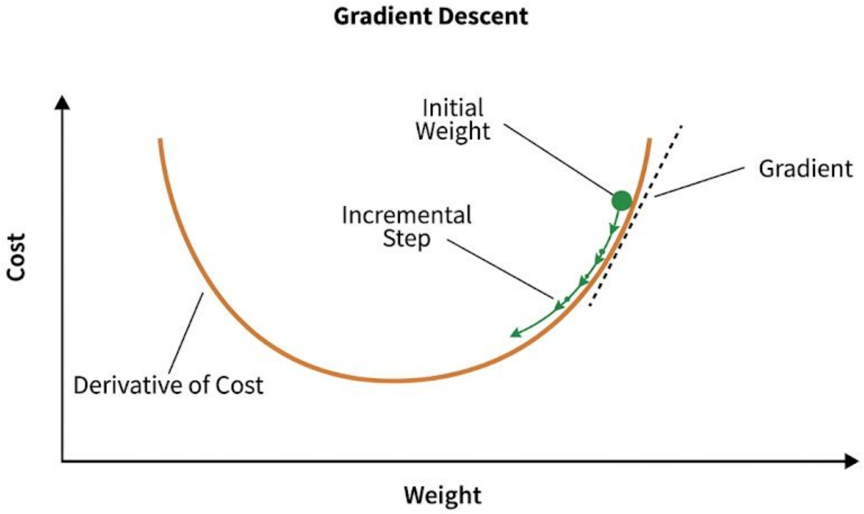
MIT: Alexander Amini, 2018 introtodeeplearning.com

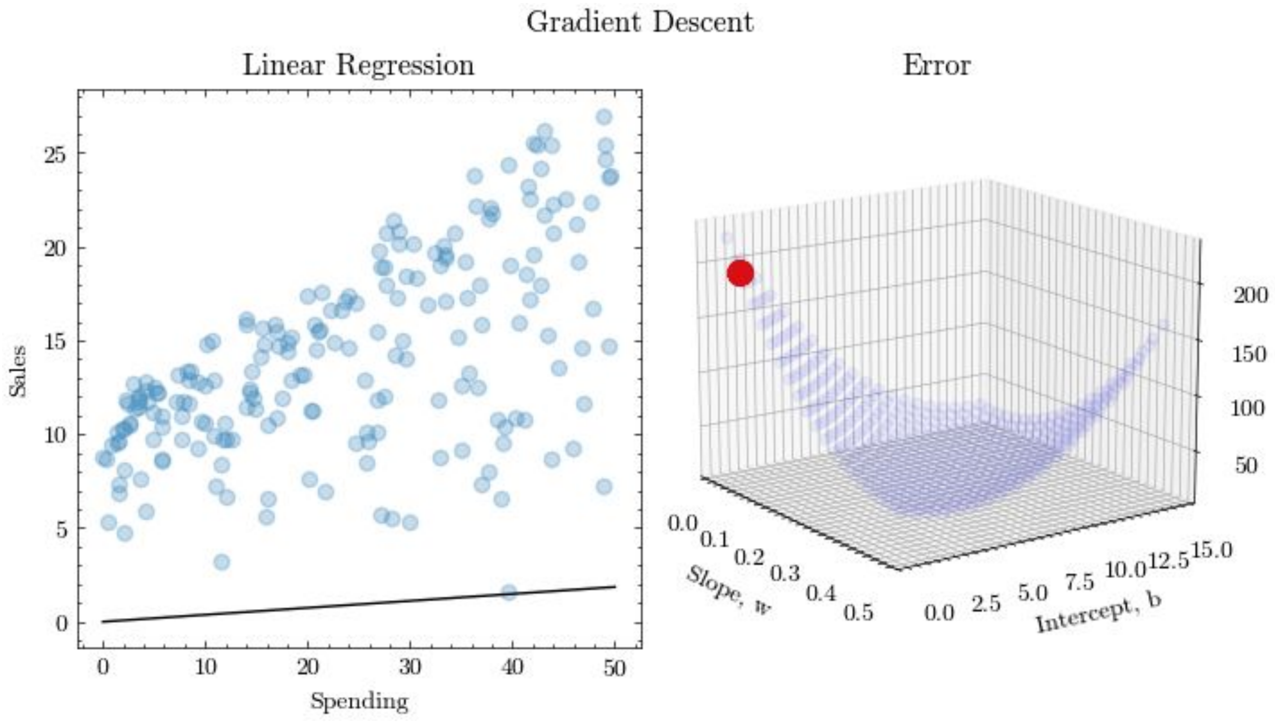

x_1
x_2
x_3
$\underbrace{w^T x + b}_{z}$
$\underbrace{\sigma(z)}_{a}$
$a = \hat{y}$


x
w
b
$\frac{\partial z}{\partial w}$
$\frac{\partial z}{\partial b}$
$z = wx^T + b$
$\frac{\partial g}{\partial z}$
$g = \frac{1}{e^{-z} + 1}$
$\frac{\partial L}{\partial g}$
$L = -\sum_i^m y \log(g) + (1 - y) \log(1 - g)$

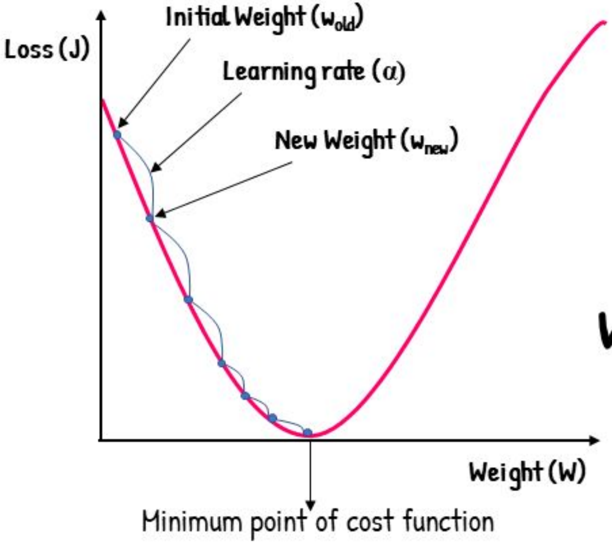

X_1
X_2
W_1
W_2
b_1
Sum
Non-Linear
Z_1
α_1
Output (α_1) of this node is the input for the next node
X_1
X_2
W_1
W_2


Cost
Learning step
Minimum
Random
initial value
w
W

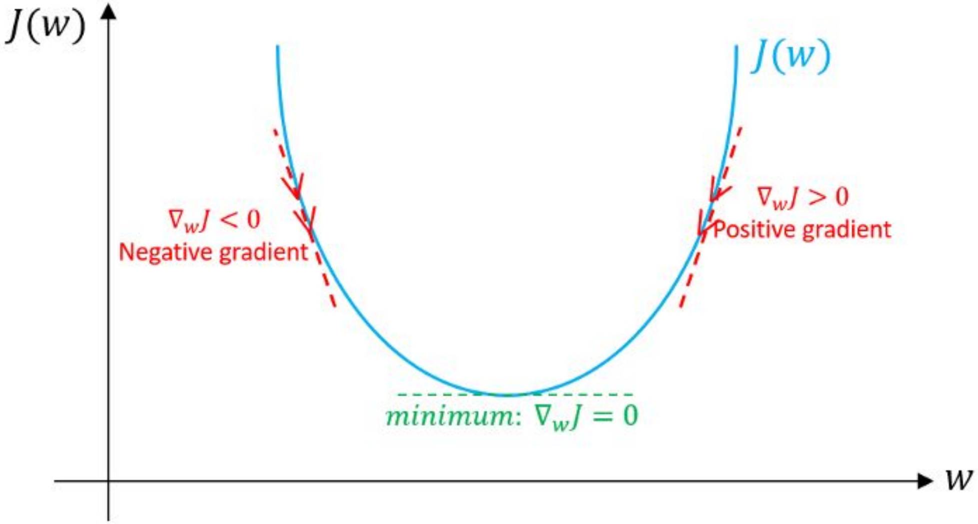

Gradient Descent
Initial
Weight
Gradient
Incremental
Step
Cost
Derivative of Cost
Weight


Gradient Descent
Linear Regression
Error
Sales
Spending
0
5
10
15
20
25
0
10
20
30
40
50
Slope, w
Intercept, b
0.0
0.1
0.2
0.3
0.4
0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
50
100
150
200

Gradient Descent

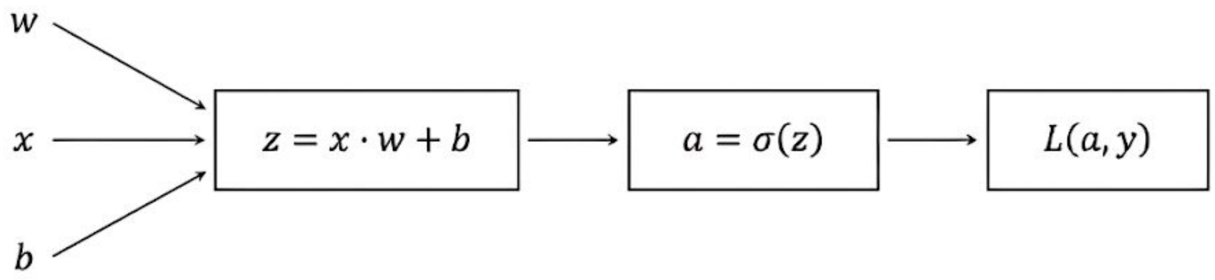


$$w_{new} = w_{old} - \alpha \frac{\delta J}{\delta w}$$

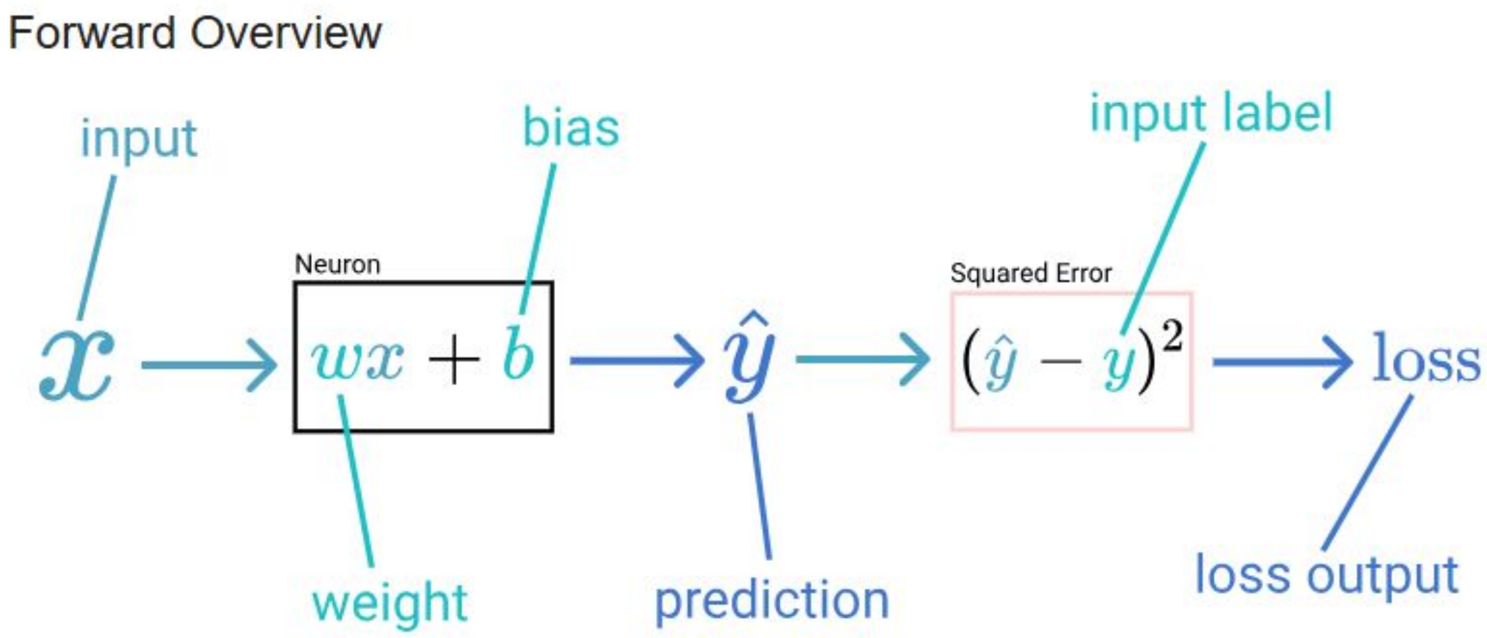


$J(w)$
$J(w)$
$\nabla_w J < 0$
Negative gradient
$\nabla_w J > 0$
Positive gradient
$minimum$: $\nabla_w J = 0$
w

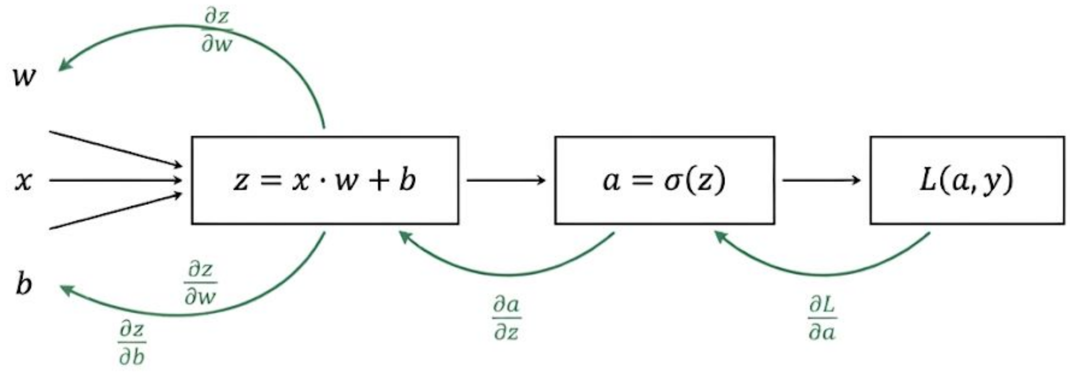
Forward Propagation



Forward Propagation



Backpropagation and Chain Rule

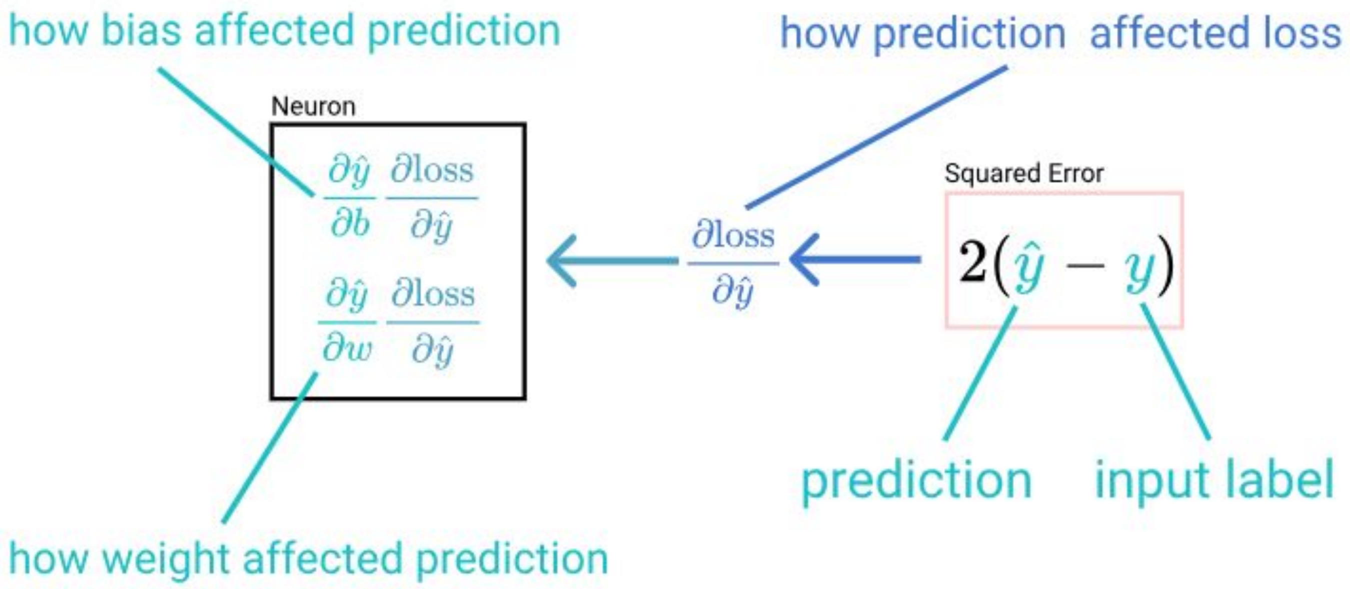


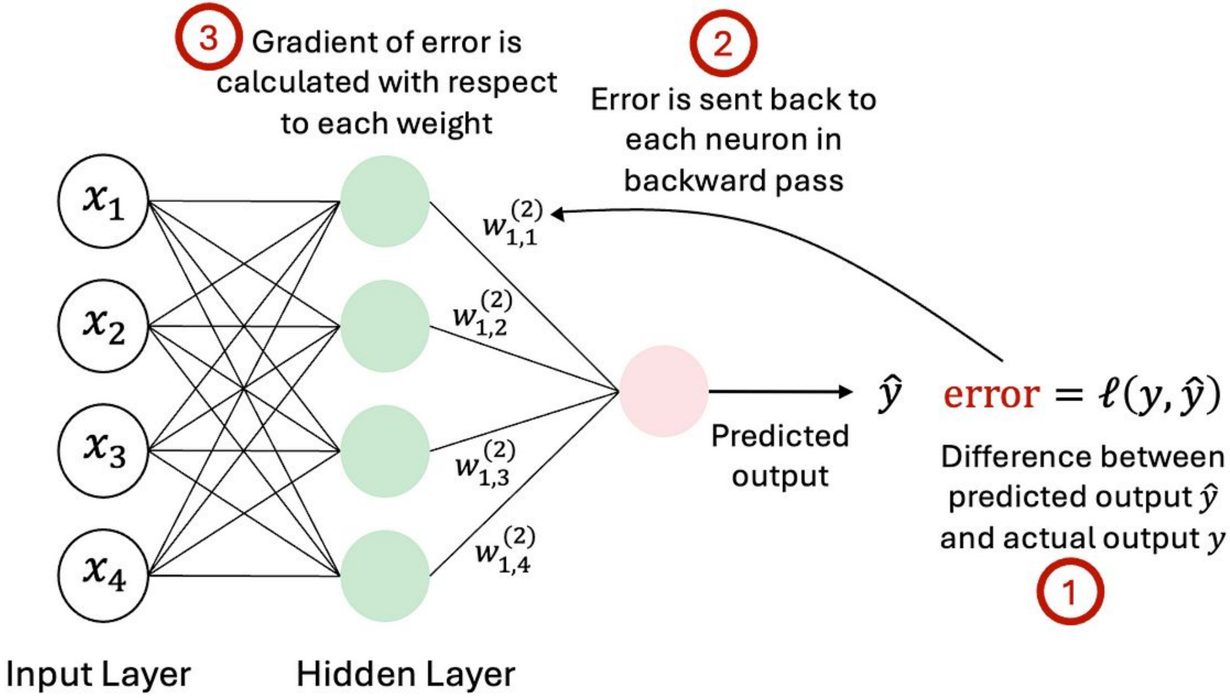
$$\frac{\partial L}{\partial w} = \frac{\partial z}{\partial w} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial L}{\partial a}$$

$$\frac{\partial L}{\partial b} = \frac{\partial z}{\partial b} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial L}{\partial a}$$

Backpropagation and Chain Rule

Backward Overview




3
Gradient of error is calculated with respect to each weight
2
Error is sent back to each neuron in backward pass
x_1
x_2
x_3
x_4
$w_{1,1}^{(2)}$
$w_{1,2}^{(2)}$
$w_{1,3}^{(2)}$
$w_{1,4}^{(2)}$
$\hat{y}$
$\text{error} = \ell(y, \hat{y})$
Predicted output
Difference between predicted output $\hat{y}$ and actual output y
1
Input Layer
Hidden Layer

W_1
W_2
W_3
W_4
W_5
W_6

Gradient Descent (1 training example)

$x1, w1, x2, w2, b \rightarrow \boxed{z = w_1x_1 + w_2x_2 + b} \rightarrow \boxed{a = g(z)} \rightarrow \boxed{L(a,y)}$

$$dz = \frac{dL}{dz} = \frac{dL}{da} * \frac{da}{dz} = (a - y)$$

$$L = -y * \log(a) + (1-y) * \log(1-a)$$

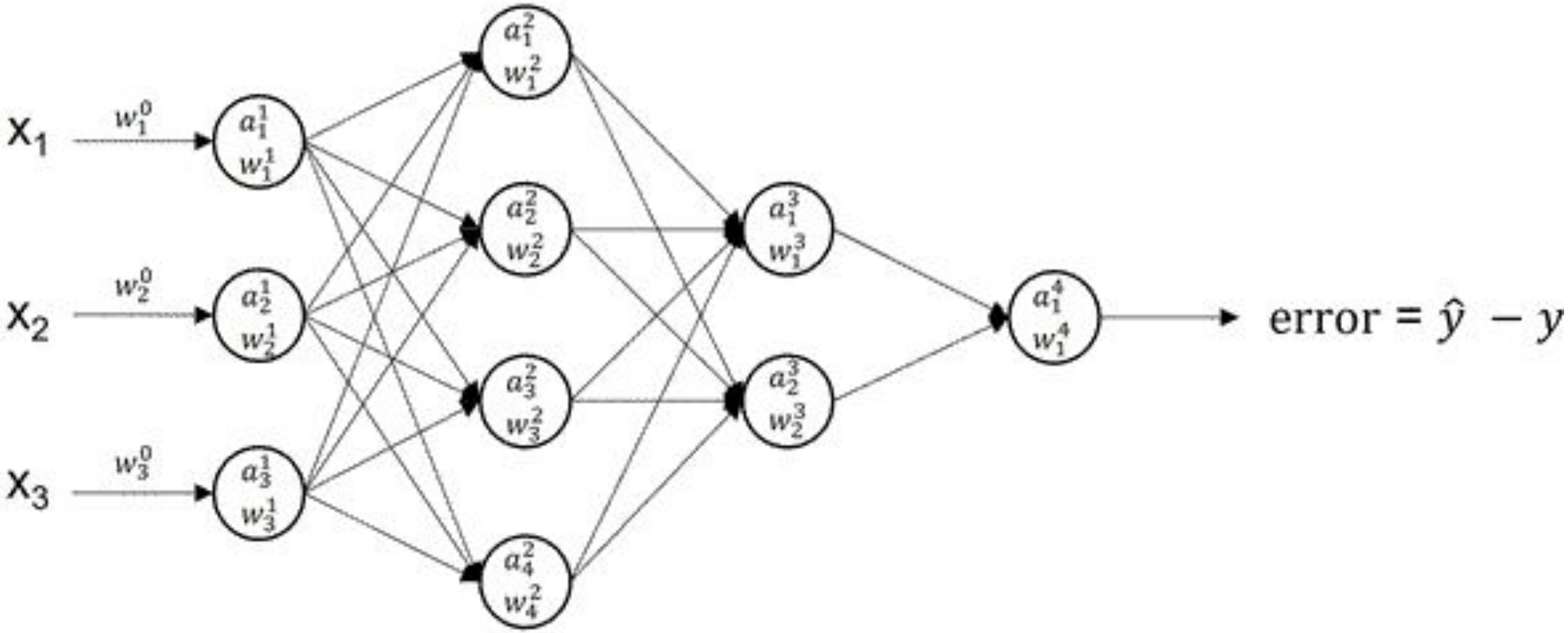
$$\frac{dL}{da} = \frac{-y}{a} + \frac{(1-y)}{(1-a)} = da$$

$$\frac{da}{dz} = a*(1-a)$$

$$dw_1: \quad \frac{dL}{dw_1} = \frac{dL}{da} * \frac{da}{dz} * \frac{dz}{dw_1} = x_1 \cdot \frac{dL}{dz}$$

$$dw_2: \quad \frac{dL}{dw_2} = \frac{dL}{da} * \frac{da}{dz} * \frac{dz}{dw_2} = x_2 \cdot \frac{dL}{dz}$$

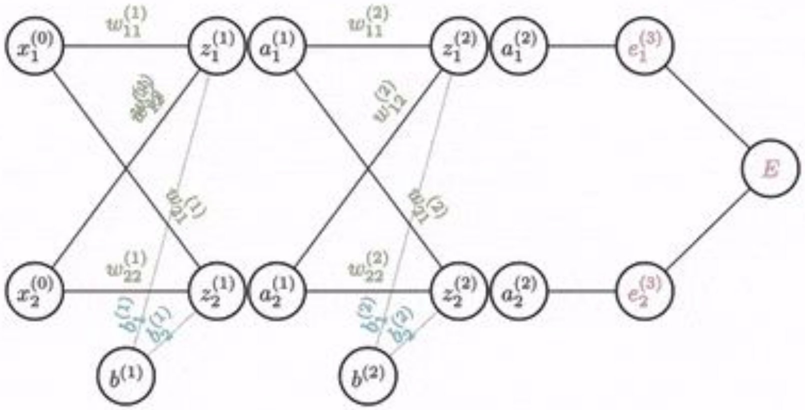
$$db: \quad \frac{dL}{db} = \frac{dL}{dz}$$



x_1
w_1^0
x_2
w_2^0
x_3
w_3^0
a_1^1 w_1^1
a_2^1 w_2^1
a_3^1 w_3^1
a_1^2 w_1^2
a_2^2 w_2^2
a_3^2 w_3^2
a_4^2 w_4^2
a_1^3 w_1^3
a_2^3 w_2^3
a_1^4 w_1^4
error = $\hat{y} - y$

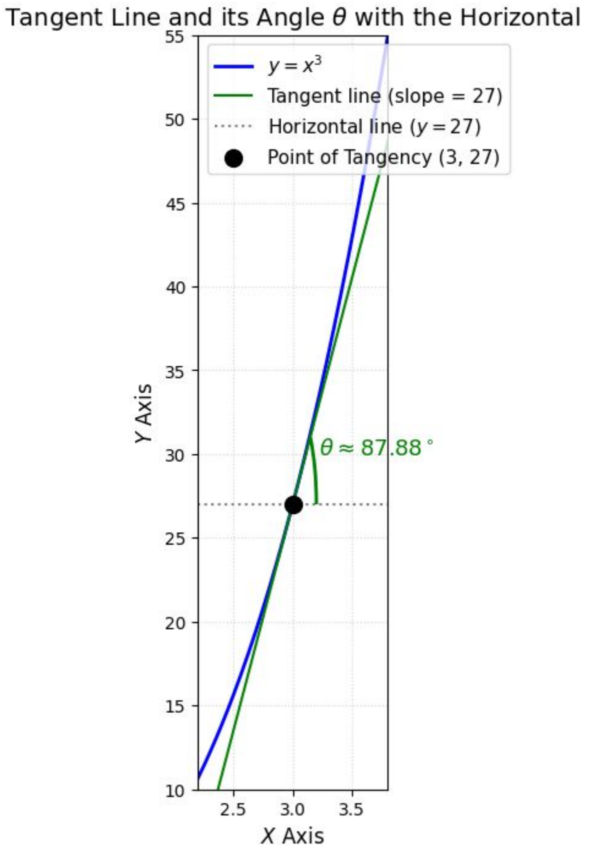
Backprop in a multilayer network

$$\frac{\partial E}{\partial w_{11}^{(1)}} = \left(\delta_1^L \frac{\partial z_1^{(2)}}{\partial a_1^{(1)}} + \delta_2^L \frac{\partial z_2^{(2)}}{\partial a_1^{(1)}} \right) \frac{\partial a_1^{(1)}}{\partial z_1^{(1)}} \frac{\partial z_1^{(1)}}{\partial w_{11}^{(1)}}$$



Training Process

1. Forward propagation resulting in an output and loss
2. Backward propagation using the chain rule to compute the gradient
3. Descend the loss by performing gradient descent



Tangent Line and its Angle θ with the Horizontal
$y = x^3$
Tangent line (slope = 27)
Horizontal line ($y = 27$)
Point of Tangency (3, 27)
$\theta \approx 87.88^\circ$
55
50
45
40
35
30
25
20
15
10
2.5
3.0
3.5
X Axis
Y Axis